A is for Annihilation

We have seen that product states can be described in terms of their occupation numbers. We now introduce a language to describe many body states, and the operators that act on them, that is adapted to this description. This method of **second quantization** represents the beginning of the field theoretic description of many body systems.

Reading: Baym (2018), Nazarov and Danon (2013).

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1 Creation and Annihilation Operators

When we discussed the elastic chain in Lecture 3, we saw that the quanta of the normal modes of the chain could be thought of as bosons. The essence of second quantization is to turn this observation around, and formally introduce **creation and annihilation operators** that add and remove bosons to a system. Subsequently, we identify these operators as components of a **quantum field**. Whether you think of the field as a physical entity – with the bosons appearing upon quantization – or as a mathematical device for representing a system of indistinguishable particles is a question of taste. The former point of view is more common in particle physics, the latter in condensed matter.

1.1 Recap: Product States

Remember that in Lecture 1 we saw that a normalized product state of N bosons occupying orthonormal states $\varphi_{\alpha_n}(\mathbf{r})$ $n = 1, \ldots, N$ is

$$\Psi^{\mathrm{S}}_{\alpha_1\alpha_2\cdots\alpha_N}(\mathbf{r}_1, \ldots, \mathbf{r}_N) = \sqrt{\frac{1}{N!\prod_\alpha N_\alpha!}} \sum_P \varphi_{\alpha_1}(\mathbf{r}_{P_1})\varphi_{\alpha_2}(\mathbf{r}_{P_2}) \cdots \varphi_{\alpha_N}(\mathbf{r}_{P_N}), \tag{1.1}$$

where the sum is over all permutations of N objects. The normalization factor involves the **occupation numbers** $\{N_\alpha\}$ giving the number of particles in state α.

More generally, let's consider the (unnormalized) symmetric product state formed from any set of single particle wavefunctions $\psi_n(\mathbf{r})$, not necessarily orthonormal

$$\Psi^{\mathrm{S}}(\mathbf{r}_1, \ldots, \mathbf{r}_N) = \frac{1}{\sqrt{N!}} \sum_P \psi_1(\mathbf{r}_{P_1})\psi_2(\mathbf{r}_{P_2}) \cdots \psi_N(\mathbf{r}_{P_N}). \tag{1.2}$$

Let's compute the inner product with another symmetric state formed from wavefunctions $\phi_n(\mathbf{r})$

$$\Phi^{\mathrm{S}}(\mathbf{r}_1, \ldots, \mathbf{r}_N) = \frac{1}{\sqrt{N!}} \sum_P \phi_1(\mathbf{r}_{P_1})\phi_2(\mathbf{r}_{P_2}) \cdots \phi_N(\mathbf{r}_{P_N}).$$

The result is

$$\langle\Psi|\Phi\rangle = \sum_P \prod_{n=1}^{N} \langle\psi_n|\phi_{P_n}\rangle = \mathrm{perm}\langle\psi_m|\phi_n\rangle, \tag{1.3}$$

where we have introduced the permanent of the matrix $\langle\psi_m|\phi_n\rangle$, which is defined just like the determinant that we met in the case of fermions, but without the minus signs.

> **Check**
>
> Satisfy yourself that Equation 1.3 agrees with Equation 1.1 in the case of orthonormal single particle wavefunctions.

General states: Fock space

We can regard a general N-particle state as a rank N tensor $\Psi_{a_1 a_2 \cdots a_N}$. We use abstract labels a_i which could index components in any basis e.g. the position basis in which we would normally write the N-body wavefunction as $\Psi(\mathbf{r}_1, \ldots \mathbf{r}_N)$. The tensor Ψ is either symmetric or antisymmetric in the labels a_i

$$\begin{aligned}\Psi^{\mathrm{S}}_{a_1 a_2 \cdots a_N} &= \Psi^{\mathrm{S}}_{(a_1 a_2 \cdots a_N)} \\ \Psi^{\mathrm{A}}_{a_1 a_2 \cdots a_N} &= \Psi^{\mathrm{A}}_{[a_1 a_2 \cdots a_N]},\end{aligned}$$

where round and square brackets denote respectively symmetrization and antisymmetrization, normalized so that they leave an already (anti)symmetric tensor unchanged

$$T_{(a_1 \cdots a_N)} = \frac{1}{N!}\sum_P T_{a_{P_1} \cdots a_{P_N}}, \qquad T_{[a_1 \cdots a_N]} = \frac{1}{N!}\sum_P (-1)^P T_{a_{P_1} \cdots a_{P_N}}.$$

Bosonic and fermionic Fock space consists of states of the form

$$|\Psi\rangle = (\Psi^{\mathrm{S/A}}, \Psi^{\mathrm{S/A}}_{a_1}, \Psi^{\mathrm{S/A}}_{a_1 a_2}, \Psi^{\mathrm{S/A}}_{a_1 a_2 a_3}, \ldots). \tag{1.4}$$

That is, each state is a combination of 0-, 1-, 2-, etc, particle states. You might find the idea of a Hilbert space of "no particles" a bit strange. This is just a bookkeeping device: it is a one-dimensional space that we call the "vacuum". We denote the only linearly independent vector in this space by $|\mathrm{VAC}\rangle$, and fix the normalization

$$\langle\mathrm{VAC}|\mathrm{VAC}\rangle = 1$$

Mathematically, we say that states of the form Equation 1.4 lie in the direct sum of the symmetric or antisymmetric subspaces of the tensor product states for N-particles for $N = 0, 1, 2, \ldots$. If you don't like that terminology, forget it!

The inner product between two Fock space vectors is given by

$$\langle \Psi | \Phi \rangle = \overline{\Psi}\Phi + \sum_{a_1} \overline{\Psi}_{a_1} \Phi_{a_1} + \sum_{a_1, a_2} \overline{\Psi}_{a_1 a_2} \Phi_{a_1 a_2} + \cdots \tag{1.5}$$

1.2 Creation and Annihilation Operators: bosonic case

There are some natural basis independent operations we can perform with symmetric tensors. Given two tensors Ψ and Φ of rank M and N respectively, we can form the symmetric product to give a symmetric tensor of rank $M + N$

$$(\Psi \odot \Phi)_{a_1 a_2 \cdots a_{M+N}} = \Psi_{(a_1 a_2 \cdots a_M} \Phi_{a_{M+1} a_{M+2} \cdots a_{M+N})}.$$

We can also contract indices to form a lower rank tensor. Thus if $M \leq N$ then contraction

$$(\bar{\Psi} \cdot \Phi)_{a_{M+1} \cdots a_N} = \bar{\Psi}_{a_1 \cdots a_M} \Phi_{a_1 a_2 \cdots a_N},$$

yields a rank $N - M$ tensor (repeated indices are summed over.) Notice that contraction preserves the symmetry of the tensors. The **annihilation operator** $a(\psi)$ is defined as the operator that acts by contracting a tensor with the one-particle state ψ:

$$(a(\psi)\Phi)_{a_2 \cdots a_N} = \sqrt{N} \overline{\psi}_{a_1} \Phi_{a_1 \cdots a_N}. \tag{1.6}$$

This action is extended to a general Fock space state Equation 1.4 by linearity. The reason for the $\sqrt{N}$ will become clear shortly. We think of the annihilation operator as "removing a particle from state ψ".

The Fock space inner product Equation 1.5 fixes the Hermitian adjoint operator $a^\dagger(\psi)$. It corresponds to the symmetric product with the one particle state ψ

$$(a^\dagger(\psi)\Phi)_{a_1 \cdots a_{N+1}} = \sqrt{N+1} \psi_{(a_1} \Phi_{a_2 \cdots a_{N+1})}. \tag{1.7}$$

We will now show that the definitions Equation 1.6 and Equation 1.7 fix the form of the commutation relations. First note that $a(\psi)$ and $a(\phi)$ commute, as well as $a^\dagger(\psi)$ and $a^\dagger(\phi)$.

$$\begin{aligned} [a^\dagger(\psi), a^\dagger(\phi)] &= 0 \\ [a(\psi), a(\phi)] &= 0. \end{aligned} \tag{1.8}$$

For the commutator of $a(\psi)$ and $a^\dagger(\phi)$ we compute the effect of acting with both on an N-particle tensor $A_{a_1\cdots a_N}$. We find

$$\begin{aligned}(a(\psi)a^\dagger(\phi)A)_{a_1\cdots a_N} &= (N+1)\overline{\psi}_b\phi_{(b}A_{a_1\cdots a_N)}\\ (a^\dagger(\phi)a(\psi)A)_{a_1\cdots a_N} &= N\overline{\psi}_b\phi_{(a_1}A_{|b|a_2\cdots a_N)}.\end{aligned}$$

Indices enclosed in vertical bars are excluded, so that $\phi_{(a_1}A_{|b|a_2\cdots a_N)}$ means symmetrization over $a_1\cdots a_N$ only. Note that in the first expression we symmetrize on $N+1$ indices $ba_1\cdots a_N$ while in the second we symmetrize on $a_1\cdots a_N$. Since A is already symmetric, the symmetrization over $ba_1\cdots a_N$ only has to choose which index sits on ϕ: b in one of the $N+1$ cases, and one of the a_j in the other N. The first expression can be written as

$$(N+1)\overline{\psi}_b\phi_{(b}A_{a_1\cdots a_N)} = \overline{\psi}_b\phi_bA_{a_1\cdots a_N} + N\overline{\psi}_b\phi_{(a_1}A_{|b|a_2\cdots a_N)},$$

This gives the commutation relation

$$[a(\psi), a^\dagger(\phi)] = \overline{\psi}_b\phi_b = \langle\psi|\phi\rangle. \tag{1.9}$$

Equation 1.8 and Equation 1.9 constitute the bosonic **canonical commutation relations**. Via Equation 1.9, this representation inherits the properties of the single particle Hilbert space.

> **Check**
>
> An N particle product state Equation 1.2 is represented in terms of creation operators as acting on the vacuum state $|\text{VAC}\rangle$ as
>
> $$\Psi^{\text{S}}(\mathbf{r}_1,\ldots,\mathbf{r}_N) \longleftrightarrow a^\dagger(\psi_1)\cdots a^\dagger(\psi_N)|\text{VAC}\rangle. \tag{1.10}$$
>
> Show that the commutation relations reproduce the correct inner product Equation 1.3.

1.3 Choosing a Basis

By choosing a basis for the single particle states, we choose a basis for the creation and annihilation operators. Denote an orthonormal basis by $\varphi_\alpha(\mathbf{r})$, with the corresponding creation and annihilation operators

$$a^\dagger(\varphi_\alpha) \equiv a^\dagger_\alpha, \quad a(\varphi_\alpha) \equiv a_\alpha. \tag{1.11}$$

Then we have

$$\left[a_\alpha, a_\beta\right] = 0, \quad \left[a_\alpha^\dagger, a_\beta^\dagger\right] = 0, \quad \left[a_\alpha, a_\beta^\dagger\right] = \delta_{\alpha\beta}.$$

The same relations describe the ladder operators of a set of independent harmonic oscillators, revealing a deep connection between these two systems. The state $\Psi^{\mathrm{S}}_{\alpha_1\alpha_2\cdots\alpha_N}$ of Equation 1.1 – a normalized state with N_α particles in state α – may be written

$$\Psi^{\mathrm{S}}_{\alpha_1\alpha_2\cdots\alpha_N}(\mathbf{r}_1, \ldots, \mathbf{r}_N) \longleftrightarrow |\mathbf{N}\rangle \equiv \prod_\alpha \frac{\left(a_\alpha^\dagger\right)^{N_\alpha}}{\sqrt{N_\alpha!}}|\mathrm{VAC}\rangle \tag{1.12}$$

The combination $\mathsf{N}_\alpha \equiv a_\alpha^\dagger a_\alpha$ is called the **number operator** for state α for obvious reasons

$$\mathsf{N}_\alpha|\mathbf{N}\rangle = N_\alpha|\mathbf{N}\rangle.$$

From Equation 1.11 it follows that

$$\begin{aligned}\left[a_\alpha, \mathsf{N}_\alpha\right] &= a_\alpha \\ \left[a_\alpha^\dagger, \mathsf{N}_\alpha\right] &= -a_\alpha^\dagger.\end{aligned}$$

You can think of the first of these as "count then destroy minus destroy then count", for example.

Suppose we want to move to a different basis of single particle states $\{|\tilde{\varphi}_\alpha\rangle\}$, corresponding to a unitary transformation

$$|\tilde{\varphi}_\alpha\rangle = \sum_\beta \langle\varphi_\beta|\tilde{\varphi}_\alpha\rangle|\varphi_\beta\rangle.$$

Using our correspondence, the one particle states with the wavefunctions $\varphi_\alpha(\mathbf{r})$ are just $a_\alpha^\dagger|\mathrm{VAC}\rangle$. So we see that the above basis transformation gives a new set of creation operators

$$\tilde{a_\alpha^\dagger} \equiv \sum_\beta \langle\varphi_\beta|\tilde{\varphi}_\alpha\rangle a_\beta^\dagger. \tag{1.13}$$

Often we will work in the basis of position eigenstates $\{|\mathbf{r}\rangle\}$. In this case the matrix elements of the unitary transformation are $\langle\varphi_\beta|\mathbf{r}\rangle = \varphi_\beta^*(\mathbf{r})$, just the complex conjugate of the wavefunction. Denoting the corresponding creation operator as $\psi^\dagger(\mathbf{r})$, Equation 1.13 becomes

$$\psi^\dagger(\mathbf{r}) \equiv \sum_\beta \varphi^*_\beta(\mathbf{r}) a^\dagger_\beta. \tag{1.14}$$

Now we can see why we chose to work with the annihilation operator rather than the creation operator. The conjugate of Equation 1.14 is

$$\psi(\mathbf{r}) \equiv \sum_\beta \varphi_\beta(\mathbf{r}) a_\beta, \tag{1.15}$$

and involves the wavefunctions $\varphi_\beta(\mathbf{r})$, rather than their conjugates. The relations satisfied by these operators are

$$\begin{aligned} [\psi(\mathbf{r}_1), \psi^\dagger(\mathbf{r}_2)] &= \delta(\mathbf{r}_1 - \mathbf{r}_2) \\ [\psi(\mathbf{r}_1), \psi(\mathbf{r}_2)] &= [\psi^\dagger(\mathbf{r}_1), \psi^\dagger(\mathbf{r}_2)] = 0. \end{aligned} \tag{1.16}$$

These operator-valued functions are called **quantum fields**. By introducing them, we have freed ourselves from many body wavefunctions. One should bear in mind that this is only a different *language* for talking about the same system: sometimes one language is better suited to the problem at hand; sometimes another.

Check

If a state $|\Psi\rangle$ has wavefunction $\Psi(x_1, \dots, x_N)$, show that the wavefunction of the state $\psi(x)|\Psi\rangle$ is the $N-1$ particle wavefunction

$$\sqrt{N}\Psi(x, x_1, \dots, x_{N-1}) \tag{1.17}$$

Hint: Show that this is true for a product state first.

Solution

Using the correspondence in Equation 1.10 we write the state that results from acting with $\psi(x)$ on a product state as

$$\psi(x)a^\dagger(\psi_1)\cdots a^\dagger(\psi_N)|\text{VAC}\rangle = \sum_m \psi_m(x) \prod_{n\neq m} a^\dagger(\psi_n)|\text{VAC}\rangle$$

where we used the commutation relation $[\psi(x), a^\dagger(\psi_n)] = \langle x|\psi_n\rangle = \psi_n(x)$. The wavefunctions appearing on the RHS on $N-1$ particle wavefunctions with arguments x_1 to x_{N-1}. Using Equation 1.10 and Equation 1.2 the RHS corresponds to

$$\frac{1}{\sqrt{(N-1)!}}\sum_m \psi_m(x)\sum_{P\in S_{N-1}}\psi_1(x_{P_1})\cdots\psi_{m-1}(x_{P_{m-1}})\psi_{m+1}(x_{P_m})\cdots\psi_N(x_{P_{N-1}})$$
$$=\frac{1}{\sqrt{(N-1)!}}\sum_{P\in S_N}\psi_1(x_{P_1})\psi_2(x_{P_2})\cdots\psi_N(x_{P_N})$$

with $x_N = x$, and S_N denotes the set of permutations of N objects (note the change from S_{N-1} to S_N between the two lines). But this just corresponds to applying the prescription Equation 1.17 to the original wavefunction Equation 1.2. Since this works for all product states, and since any state is a superposition of product states, it works more generally.

As an example, let our original basis be the eigenbasis of the free particle Hamiltonian $H = \frac{\mathbf{p}^2}{2m}$ with periodic boundary conditions

$$|\mathbf{k}\rangle = \frac{\exp(i\mathbf{k}\cdot\mathbf{r})}{\sqrt{V}}, \quad \mathbf{k} = 2\pi\left(\frac{n_x}{L_x}, \frac{n_y}{L_y}, \frac{n_z}{L_z}\right), \quad n_{x,y,z} \text{ integer},$$

with $V = L_x L_y L_z$. The matrix elements of the transformation between this original basis and the position basis $\{|\mathbf{r}\rangle\}$ are $\langle\mathbf{k}|\mathbf{r}\rangle = \exp(-i\mathbf{k}\cdot\mathbf{r})/\sqrt{V}$, so we have

$$\psi^\dagger(\mathbf{r}) \equiv \frac{1}{\sqrt{V}}\sum_{\mathbf{k}}\exp(-i\mathbf{k}\cdot\mathbf{r})a^\dagger_{\mathbf{k}},$$

and similarly

$$\psi(\mathbf{r}) \equiv \frac{1}{\sqrt{V}}\sum_{\mathbf{k}}\exp(i\mathbf{k}\cdot\mathbf{r})a_{\mathbf{k}}.$$

Check

What is the wavefunction of the two-particle state

$$\sum_{\mathbf{k}} c_k a^\dagger_{\mathbf{k}} a^\dagger_{-\mathbf{k}}|\text{VAC}\rangle?$$

1.4 The Case of Fermions

We want a similar formalism for fermions, which is a bit trickier on account of those minus signs. We seek a representation of the product state

$$\Psi^{\mathrm{A}}(\mathbf{r}_1, \ldots, \mathbf{r}_N) = \frac{1}{\sqrt{N!}} \sum_P (-1)^P \psi_1(\mathbf{r}_{P_1}) \psi_2(\mathbf{r}_{P_2}) \cdots \psi_N(\mathbf{r}_{P_N}).$$

where P is the signature of the permutation. Note that the overall sign is fixed here by the labelling of the states ψ_j. If we want to make the identification

$$\Psi^{\mathrm{A}}(\mathbf{r}_1, \ldots, \mathbf{r}_N) \longleftrightarrow a^\dagger(\psi_1) \cdots a^\dagger(\psi_N)|\mathrm{VAC}\rangle, \tag{1.18}$$

Then antisymmetry dictates

$$\{a^\dagger(\psi), a^\dagger(\phi)\} = 0,$$

for any states $\psi(\mathbf{r})$ and $\phi(\mathbf{r})$, where $\{A, B\} \equiv AB + BA$ is called the **anticommutator**. Similarly

$$\{a(\psi), a(\phi)\} = 0.$$

The form of $\{a(\psi), a^\dagger(\phi)\}$ can be deduced from the inner product between product states

$$\langle \Psi | \Phi \rangle = \sum_P (-1)^P \prod_{n=1}^N \langle \psi_n | \phi_{P_n} \rangle = \det \langle \psi_m | \phi_n \rangle,$$

which can be obtained using the RHS of Equation 1.18 if we demand

$$\{a(\psi), a^\dagger(\phi)\} = \langle \psi | \phi \rangle.$$

> **Check**
>
> Check this.

Introducing field operators in the position basis as before leads to a system of operators satisfying the anticommutation relations

$$\begin{aligned} \{\psi(\mathbf{r}_1), \psi^\dagger(\mathbf{r}_2)\} &= \delta(\mathbf{r}_1 - \mathbf{r}_2) \\ \{\psi(\mathbf{r}_1), \psi(\mathbf{r}_2)\} &= \{\psi^\dagger(\mathbf{r}_1), \psi^\dagger(\mathbf{r}_2)\} = 0. \end{aligned} \tag{1.19}$$

Check

Show that in the fermionic case the wavefunction of the state $\psi(x)|\Psi\rangle$ is still given by Equation 1.17.

2 Discussion: How Do a_α and $a_\alpha^\dagger$ *Look*?

Check

Think about the form that the operators a_α, $a_\alpha^\dagger$ take in the basis of product states. Start with one state φ_α. What's the matrix form of $a_\alpha^\dagger$ in terms of states $|N_\alpha\rangle$? Now consider two states. Can you see how the commutation and anticommutation relations can be satisfied?

3 Representation of Operators

3.1 One Particle Operators

Suppose we now have some operator A that acts on the single particle states. We denote the action on N particle states as $\hat{A}$

$$\hat{A} = \sum_{j=1}^{N} A_j, \tag{3.1}$$

where the subscript denotes that the operator acts on the position label $\mathbf{r}_j$ of the jth particle. The example that we have met many times already is the Hamiltonian for noninteracting particles

$$\hat{H} = \sum_{j=1}^{N} H_j = \sum_{j=1}^{N} \left[-\frac{\nabla_j^2}{2m} + V(\mathbf{r}_j) \right]. \tag{3.2}$$

Operators of this type are known as **one particle operators**. We want to use our creation and annihilation operators to represent these operators.

First, the action of the operator A on one of the basis states $|\varphi_\beta\rangle$ can be written in terms of the matrix elements $A_{\alpha\beta} = \langle\varphi_\alpha|A|\varphi_\beta\rangle$ by inserting a resolution of the identity $\sum_\alpha |\alpha\rangle\langle\alpha| = 1$ on the right

$$A|\varphi_\beta\rangle = \sum_\alpha |\varphi_\alpha\rangle\langle\varphi_\alpha|A|\varphi_\beta\rangle = \sum_\alpha A_{\alpha\beta}|\varphi_\alpha\rangle.$$

The same thing may be expressed slightly differently by inserting resolutions of the identity on the left *and* right

$$A = \sum_{\alpha\beta} |\varphi_\alpha\rangle\langle\varphi_\alpha|A|\varphi_\beta\rangle\langle\varphi_\beta| = \sum_{\alpha\beta} A_{\alpha\beta}|\varphi_\alpha\rangle\langle\varphi_\beta|.$$

Therefore, the action of $\hat{A}$ on the product state $|\Psi^{\text{S}}_{\alpha_1\alpha_2\cdots\alpha_N}\rangle$ is

$$\hat{A}|\Psi^{\text{S}}_{\beta_1\beta_2\cdots\beta_N}\rangle = \sum_\alpha \left[A_{\alpha\beta_1}|\Psi^{\text{S}}_{\alpha\beta_2\cdots\beta_N}\rangle + A_{\alpha\beta_2}|\Psi^{\text{S}}_{\beta_1\alpha\cdots\beta_N}\rangle + \cdots A_{\alpha\beta_N}|\Psi^{\text{S}}_{\beta_1\beta_2\cdots\alpha}\rangle \right] \tag{3.3}$$

Now we argue that the same job is done by

$$\hat{A} = \sum_{\alpha\beta} A_{\alpha\beta} a_\alpha^\dagger a_\beta$$

acting on the state $|\mathbf{N}\rangle$. Indeed, we have

$$\left[a_\alpha^\dagger a_\beta, a_\gamma^\dagger\right] = a_\alpha^\dagger \delta_{\beta\gamma} \tag{3.4}$$

By commuting $a_\alpha^\dagger a_\beta$ though each of the creation operators in

$$|\mathbf{N}\rangle \equiv \prod_\alpha \frac{\left(a_\alpha^\dagger\right)^{N_\alpha}}{\sqrt{N_\alpha!}} |\mathrm{VAC}\rangle,$$

we reproduce the action Equation 3.3. As a simple example,

$$\begin{aligned} \hat{A}\, a_\beta^\dagger |\mathrm{VAC}\rangle &= \left(\left[\hat{A}, a_\beta^\dagger\right] + a_\beta^\dagger\, \hat{A}\right) |\mathrm{VAC}\rangle \\ &= \sum_\alpha A_{\alpha\beta} a_\alpha^\dagger |\mathrm{VAC}\rangle. \end{aligned}$$

The second quantized form of $\hat{A}$ allows us to find the matrix element $\langle \mathbf{N}|\hat{A}|\mathbf{N}'\rangle$ between product states made of orthonormal single particle states. This vanishes unless $\mathbf{N}$ and $\mathbf{N}'$ differ by the movement of of one particle. If $N_\beta = N'_\beta - 1$ and $N_\alpha = N'_\alpha + 1$ we have

$$\langle \mathbf{N}|\hat{A}|\mathbf{N}'\rangle = A_{\alpha\beta}\sqrt{N_\alpha N'_\beta}. \tag{3.5}$$

> **Check**
>
> This formula is not so easy to work out in the first quantized representation. Try it!

We worked this out for the bosonic case. However, you can check that Equation 3.4, the basic relation that makes it all work, is also valid for fermions (as a commutation relation, note!). So $\hat{A}$ has the same form for fermions.

Notice that $\hat{A}$ *looks* formally like the expectation value of A in a single particle state $\sum_\alpha a_\alpha |\varphi_\alpha\rangle$. The difference, of course, is that the a_α in $\hat{A}$ are operators, so that the order is important, while those in the preceding expression are amplitudes. The replacement of amplitudes, or wavefunctions, by operators is the origin of the rather clumsy name **second quantization**, which is traditionally introduced with the caveat that what we are doing is not in any way 'more quantum' than before.

To repeat the above prescription for emphasis: *A one particle operator $\hat{A}$ has a second quantized representation formally identical to the expectation value of its single particle counterpart A.*

Check

Another way to justify the second quantized form of the one particle operator is to start from the expression

$$\hat{A} = \int \psi^\dagger(\mathbf{r}) \left[A\psi(\mathbf{r})\right] d\mathbf{r}$$

and take matrix elements with general N-particle state $|\Psi\rangle$, $|\Phi\rangle$. Using Equation 1.17 we find

$$\langle\Psi|\hat{A}|\Phi\rangle \longleftrightarrow N \int \Psi^*(\mathbf{r}_1, \ldots \mathbf{r}_N) \left[A_1 \Phi(\mathbf{r}_1, \ldots \mathbf{r}_N)\right] d\mathbf{r}_1 \cdots d\mathbf{r}_N.$$

By the symmetry or antisymmetry of the wavefunctions this is the same as taking matrix elements of the first quantized form Equation 3.1.

This probably all looks a bit abstract, so let's turn to a one particle operator that we have already met, namely the noninteracting Hamiltonian in Equation 3.2. According to the above prescription, this should have the second quantized form

$$\hat{H} \equiv \sum_{\alpha,\beta} \langle\varphi_\alpha| H |\varphi_\beta\rangle a_\alpha^\dagger a_\beta,$$

where H is the single particle Hamiltonian $H = -\frac{1}{2m}\nabla_i^2 + V(\mathbf{r_i})$. This takes on a very simple form if the basis $|\varphi_\alpha\rangle$ is just the eigenbasis of this Hamiltonian, in which case $\langle\varphi_\alpha| H |\varphi_\beta\rangle = E_\alpha \delta_{\alpha\beta}$ and

$$\hat{H} \equiv \sum_\alpha E_\alpha a_\alpha^\dagger a_\alpha = \sum_\alpha E_\alpha \mathsf{N}_\alpha. \tag{3.6}$$

Evidently this is correct: the eigenstates of this operator are just the N particle basis states $|\Psi^{S/A}_{\alpha_1\alpha_2\cdots\alpha_N}\rangle$, with eigenvalues $\sum_\alpha N_\alpha E_\alpha$.

Alternatively, we can look at things in the position basis. By recalling how the expectation value of the Hamiltonian looks in this basis, we come up with

$$\begin{aligned}\hat{H} &= \int d\mathbf{r} \left[-\frac{1}{2m}\psi^\dagger(\mathbf{r})\nabla^2\psi(\mathbf{r}) + V(\mathbf{r})\psi^\dagger(\mathbf{r})\psi(\mathbf{r})\right] \\ &= \int d\mathbf{r} \left[\frac{1}{2m}\nabla\psi^\dagger(\mathbf{r})\cdot\nabla\psi(\mathbf{r}) + V(\mathbf{r})\psi^\dagger(\mathbf{r})\psi(\mathbf{r})\right],\end{aligned}$$

where in the second line we have integrated by parts, assuming that boundary terms at infinity vanish. The equality of Equation 3.7 and Equation 3.6 may be seen by using Equation 1.15.

The Heisenberg equation of motion for $\psi(\mathbf{r},t) = e^{i\hat{H}t}\psi(\mathbf{r})e^{-i\hat{H}t}$ with Hamiltonian Equation 3.7 is

$$\begin{aligned} i\partial_t\psi(\mathbf{r},t) &= -\left[\hat{H},\psi(\mathbf{r},t)\right] \\ &= -\frac{1}{2m}\nabla^2\psi(\mathbf{r},t) + V(\mathbf{r})\psi(\mathbf{r},t), \end{aligned}$$

which is just the time dependent Schrödinger equation!

As a second example, consider the particle density. This is not something that one encounters very often in few particle quantum mechanics, but is obviously an observable of interest in a extended system of many particles. The one particle operator for the density at $\mathbf{x}$ is

$$\rho(\mathbf{x}) \equiv \delta(\mathbf{x}-\mathbf{r}).$$

This may look like a rather strange definition, but its expectation value on a single particle state $\varphi(\mathbf{r})$ is just $\rho(\mathbf{x}) = |\varphi(\mathbf{x})|^2$, which is just the probability to find the particle at $\mathbf{x}$. Following our prescription, the second quantized form of the operator is then

$$\hat{\rho}(\mathbf{x}) \equiv \psi^\dagger(\mathbf{x})\psi(\mathbf{x}).$$

As a check, integrating over position should give the total number of particles

$$\hat{N} = \int d\mathbf{x}\,\psi^\dagger(\mathbf{x})\psi(\mathbf{x}) = \sum_\alpha a_\alpha^\dagger a_a = \sum_\alpha \mathsf{N}_\alpha,$$

as it does! Another useful thing to know is the expectation value of the density on a basis state $|N_0, N_1 ...\rangle$

$$\langle N_0, N_1 ...|\hat{\rho}(\mathbf{r})|N_0, N_1 ...\rangle = \sum_\alpha N_\alpha \left|\varphi_\alpha(\mathbf{r})\right|^2.$$

which is most easily proved by substituting the representation Equation 1.15. This seems like a very reasonable generalization of the single particle result: the density is given by sum of the probability densities in each of the constituent single particle state, weighted by the occupancy of the state. Note that the symmetry of the states played no role here.

As a final example of a one particle operator, the particle current has the second quantized form

$$\hat{\mathbf{j}}(\mathbf{r}) = -i\frac{1}{2m}\left[\psi^\dagger(\mathbf{r})\left(\nabla\psi(\mathbf{r})\right) - \left(\nabla\psi^\dagger(\mathbf{r})\right)\psi(\mathbf{r})\right].$$

Often we consider the Fourier components of the density or current

$$\hat{\rho}_{\mathbf{q}} \equiv \int d\mathbf{r}\,\hat{\rho}(\mathbf{r})e^{-i\mathbf{q}\cdot\mathbf{r}} = \sum_{\mathbf{k}} a^\dagger_{\mathbf{k}-\mathbf{q}}a_{\mathbf{k}}$$
$$\hat{\mathbf{j}}_{\mathbf{q}} \equiv \int d\mathbf{r}\,\hat{\mathbf{j}}(\mathbf{r})e^{-i\mathbf{q}\cdot\mathbf{r}} = \frac{1}{m}\sum_{\mathbf{k}}(\mathbf{k}-\mathbf{q}/2)a^\dagger_{\mathbf{k}-\mathbf{q}}a_{\mathbf{k}}.$$

The $\mathbf{q} = 0$ modes are just the total particle number and $\frac{1}{m}$ times the total momentum, respectively.

In Lecture 1 we introduced the **single particle density matrix**

$$g(\mathbf{r},\mathbf{r}') \equiv N\int d\mathbf{r}_2\cdots d\mathbf{r}_N\,\Psi^*(\mathbf{r},\mathbf{r}_2,\ldots,\mathbf{r}_N)\Psi(\mathbf{r}',\mathbf{r}_2,\ldots,\mathbf{r}_N).$$

Check

Show that this can be written in terms of our field operators as

$$g(\mathbf{r},\mathbf{r}') = \langle\Psi|\psi^\dagger(\mathbf{r})\psi(\mathbf{r}')|\Psi\rangle.$$

Notice that $g(\mathbf{r},\mathbf{r}) = \langle\Psi|\hat{\rho}(\mathbf{r})|\Psi\rangle$. A slight generalization of the above calculation for the density gives for the state $|\mathbf{N}\rangle$

$$g(\mathbf{r},\mathbf{r}') = \sum_\alpha N_\alpha\varphi^*_\alpha(\mathbf{r})\varphi_\alpha(\mathbf{r}').$$

In Lecture 1 we evaluated $g(x,y)$ for the ground state of a 1D Fermi gas using the Slater determinant form of the wavefunction. For variety, let's calculate it using our new method for the 3D Fermi gas. Recall that in this case $N_{\mathbf{k}} = 1$ for $|\mathbf{k}| < k_F$, and 0 otherwise. Thus the ground state can be written

$$|\text{Fermi sea}\rangle = \prod_{|\mathbf{k}|<k_F} a^\dagger_{\mathbf{k}}|\text{VAC}\rangle$$

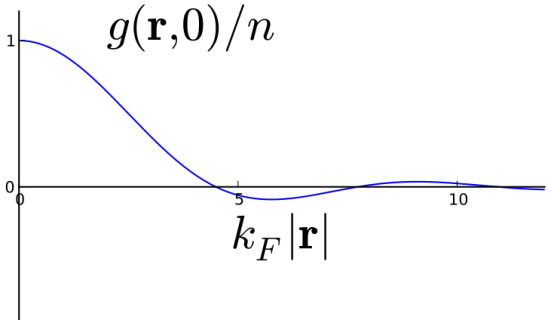
Check

Show that

$$g(\mathbf{r},\mathbf{r}') = \frac{1}{V}\sum_{|\mathbf{k}|<k_F} e^{i\mathbf{k}\cdot(\mathbf{r}'-\mathbf{r})} = \int_{|\mathbf{k}|<k_F} \frac{d\mathbf{k}}{(2\pi)^3}\, e^{i\mathbf{k}\cdot(\mathbf{r}'-\mathbf{r})}$$
$$= \frac{k_F^3}{2\pi^2}\left[\frac{\sin\left(k_F|\mathbf{r}'-\mathbf{r}|\right)}{(k_F|\mathbf{r}'-\mathbf{r}|)^3} - \frac{\cos\left(k_F|\mathbf{r}'-\mathbf{r}|\right)}{(k_F|\mathbf{r}'-\mathbf{r}|)^2}\right].$$

Note that $g(\mathbf{r},\mathbf{r}) = \frac{k_F^3}{6\pi^2} = n$, as it should. Also, $g(\mathbf{r},\mathbf{r}') \to 0$ as $|\mathbf{r}-\mathbf{r}'| \to \infty$.

Contrast this calculation with the wavefunction version.



3.2 Two Particle Operators

A **two particle operator** acts pairwise on the particles

$$\hat{B} = \sum_{j<k} B_{jk}. \tag{3.7}$$

Note that $B_{jk} = B_{kj}$ for indistinguishable particles, and that $j = k$ is excluded. The most important two particle operator that we will encounter describes interactions between pairs of particles, usually of the form

$$\hat{H}_{\text{int.}} = \sum_{j<k} U(\mathbf{r}_j - \mathbf{r}_k). \tag{3.8}$$

Similar to the single particle case, the action of $\hat{B}$ on a two particle product state $|\varphi_\alpha\rangle_1|\varphi_\beta\rangle_2$ can be expressed in terms of the matrix elements

$$B_{\alpha\beta,\gamma\delta} = \langle\varphi_\alpha|\langle\varphi_\beta|B_{12}|\varphi_\gamma\rangle|\varphi_\delta\rangle$$
$$= \int d\mathbf{r}d\mathbf{r}'\varphi_\alpha^*(\mathbf{r})\varphi_\beta^*(\mathbf{r}')B_{12}\varphi_\gamma(\mathbf{r})\varphi_\delta(\mathbf{r}')$$

Note that since $B_{12} = B_{21}$ we have $B_{\alpha\beta,\gamma\delta} = B_{\beta\alpha,\delta\gamma}$. In the first quantized representation we have

$$B_{12} = \sum_{\alpha\beta\gamma\delta} B_{\alpha\beta,\gamma\delta} |\varphi_\alpha\rangle|\varphi_\beta\rangle\langle\varphi_\gamma|\langle\varphi_\delta|.$$

Now we'll argue that the second quantized representation of $\hat{B}$ is

$$\hat{B} = \frac{1}{2} \sum_{\alpha\beta\gamma\delta} B_{\alpha\beta,\gamma\delta} a_\alpha^\dagger a_\beta^\dagger a_\delta a_\gamma. \tag{3.9}$$

(Note the order, which is important for fermions!).

Check

Check that Equation 3.9 has the correct action on product states, starting with the two particle state

$$a_\gamma^\dagger a_\delta^\dagger |\text{VAC}\rangle \longleftrightarrow \frac{1}{\sqrt{2}} \left[\varphi_\gamma(\mathbf{r}_1)\varphi_\delta(\mathbf{r}_2) \pm \varphi_\delta(\mathbf{r}_1)\varphi_\gamma(\mathbf{r}_2)\right]$$

For example, in the first quantized picture we have

$$B_{12} \frac{1}{\sqrt{2}} \left[\varphi_\gamma(\mathbf{r}_1)\varphi_\delta(\mathbf{r}_2) \pm \varphi_\delta(\mathbf{r}_1)\varphi_\gamma(\mathbf{r}_2)\right] = \frac{1}{\sqrt{2}} \sum_{\alpha\beta} \left[B_{\alpha\beta,\gamma\delta} \pm B_{\alpha\beta,\delta\gamma}\right] \varphi_\alpha(\mathbf{r}_1)\varphi_\beta(\mathbf{r}_2).$$

Note that this is (anti-)symmetric for bosons (fermions) by virtue of $B_{\alpha\beta,\gamma\delta} = B_{\beta\alpha,\delta\gamma}$. In second quantized notation the resulting state is written

$$\frac{1}{\sqrt{2}} \sum_{\alpha\beta} \left[B_{\alpha\beta,\gamma\delta} \pm B_{\alpha\beta,\delta\gamma}\right] \varphi_\alpha(\mathbf{r}_1)\varphi_\beta(\mathbf{r}_2) \longleftrightarrow \sum_{\alpha\beta} B_{\alpha\beta} a_\alpha^\dagger a_\beta^\dagger |\text{VAC}\rangle$$

Which is the same as we get if we act with Equation 3.9

$$\left(\frac{1}{2} \sum_{\alpha\beta cd} B_{\alpha\beta,cd} a_\alpha^\dagger a_\beta^\dagger a_d a_c\right) a_\gamma^\dagger a_\delta^\dagger |\text{VAC}\rangle = \sum_{\alpha\beta} B_{\alpha\beta} a_\alpha^\dagger a_\beta^\dagger |\text{VAC}\rangle.$$

Make sure to trace down factor of 2!

The analog of the formula Equation 3.5 for the matrix elements of a two particle operator between two product states is

$$\langle \mathbf{N}|\hat{B}|\mathbf{N}'\rangle = \left[B_{\alpha\beta,\gamma\delta} \pm B_{\alpha\beta,\delta\gamma}\right] \sqrt{N_\alpha N_\beta N'_\gamma N'_\delta},$$

with $N_{\gamma,\delta} = N'_{\gamma,\delta} - 1$ and $N_{\alpha,\beta} = N'_{\alpha,\beta} + 1$.

> **Check**
>
> Strictly this formula is not correct when $\gamma = \delta$ or $\alpha = \beta$. In the former case $N'_\gamma N'_\delta \to N'_\gamma(N'_\gamma - 1)$ and in the latter $N_\alpha N_\beta \to N_\alpha(N_\alpha - 1)$. In the thermodynamic limit these terms usually make a vanishing contribution when sums are replaced with integrals. There are exceptions: when a finite fraction of the particles are in one state (which occurs for Bose—Einstein condensates, for example). In those cases we end up neglecting $N_\alpha - 1$ relative to N_α, however!

Returning to the interaction Equation 3.8, we can express this operator in the position basis as

$$\hat{H}_{\text{int.}} = \frac{1}{2}\int d\mathbf{r}_1 d\mathbf{r}_2\, U(\mathbf{r}_1 - \mathbf{r}_2)\psi^\dagger(\mathbf{r}_1)\psi^\dagger(\mathbf{r}_2)\psi(\mathbf{r}_2)\psi(\mathbf{r}_1).$$

Remembering that $\rho(\mathbf{r}) = \psi^\dagger(\mathbf{r})\psi(\mathbf{r})$, this is *almost* the same as

$$\hat{H}_{\text{int.}} = \frac{1}{2}\int d\mathbf{r}_1 d\mathbf{r}_2\, U(\mathbf{r}_1 - \mathbf{r}_2)\rho(\mathbf{r}_1)\rho(\mathbf{r}_2). \tag{3.10}$$

The difference lies in the order of the operators, which prevents a particle from interacting with itself. Operators in which all annihilation operators stand to the right of all creation operators are said to be **normal ordered**.

> **Check**
>
> Check that Equation 3.10 has a zero expectation value in a 1 particle state.

As an example, let's write the Hamiltonian of interacting bosons that we met in Lecture 1 in our new formalism.

$$H = \int dx \left[\frac{1}{2}\partial_x\psi^\dagger(x)\partial_x\psi(x) + \frac{c}{2}\psi^\dagger(x)\psi^\dagger(x)\psi(x)\psi(x)\right],$$

where $\psi(x)$, $\psi^\dagger(x)$ satisfy the canonical bosonic commutation relations Equation 1.16.

We've succeeded in translating an interacting many body Hamiltonian into a quantum field theory. But we shouldn't feel too smug. As I said earlier, this is a change of *language*, and its usefulness is not guaranteed. The same warning applies to alternative formulations of the same problem: as a functional integral, for example. In the case of the Lieb–Liniger model, the exact solution was best presented in the first quantized, many body wavefunction framework. Ditto for the Laughlin states of the fractional quantum Hall effect. However, in approximate treatments of many body problems that we *can't* solve exactly, like the Bose gas in higher dimensions that we'll discuss in Lecture 8, the second quantized framework is often a good starting point.

4 Appendices

4.1 Derivation of the anticommutation relations for fermions

The fermionic case proceeds similarly, except that we have to be a bit cautious about signs. Two antisymmetric tensors of rank M and N are combined into an antisymmetric tensor of rank $M+N$ using the antisymmetric (or *wedge*) product

$$(\Psi \wedge \Phi)_{a_1 a_2 \cdots a_{M+N}} = \Psi_{[a_1 a_2 \cdots a_M} \Phi_{a_{M+1} a_{M+2} \cdots a_{M+N}]}.$$

Unlike the symmetric product, the order now matters: $\Psi \wedge \Phi = (-1)^{MN}\, \Phi \wedge \Psi$. Contraction also preserves antisymmetry, but which index we contract affects the sign, since moving an index from the last position to the first costs a factor $(-1)^{N-1}$. We define the creation operator to place the new particle in the *first* slot

$$(a^\dagger(\psi)\Phi)_{a_1 \cdots a_{N+1}} = \sqrt{N+1}\, \psi_{[a_1} \Phi_{a_2 \cdots a_{N+1}]}, \tag{4.1}$$

so that $a^\dagger(\psi_1) a^\dagger(\psi_2) \cdots a^\dagger(\psi_N)|\mathrm{VAC}\rangle$ is the antisymmetrized product state Equation 1.18, with ψ_1 in the first position and the factors of $\sqrt{N}$ combining with the $1/N!$ in the antisymmetrizer to give the normalization $1/\sqrt{N!}$. The annihilation operator must then contract the *first* index

$$(a(\psi)\Phi)_{a_2 \cdots a_N} = \sqrt{N}\, \overline{\psi}_b \Phi_{b a_2 \cdots a_N}. \tag{4.2}$$

These are Hermitian conjugates with respect to the inner product Equation 1.5. For an $(N-1)$-particle state X we have

$$\langle a^\dagger(\psi) X | \Phi \rangle = \sqrt{N}\, \overline{\psi}_{[b} \overline{X}_{a_2 \cdots a_N]} \Phi_{b a_2 \cdots a_N} = \sqrt{N}\, \overline{X}_{a_2 \cdots a_N} \overline{\psi}_b \Phi_{b a_2 \cdots a_N} = \langle X | a(\psi)\Phi \rangle,$$

because the antisymmetrizer can be moved onto Φ, which it leaves unchanged. Had we contracted the last index in Equation 4.2, this would only hold up to the sign $(-1)^{N-1}$.

Applying two annihilation operators gives

$$(a(\psi)a(\phi)\Phi)_{a_3 \cdots a_N} = \sqrt{N(N-1)}\, \overline{\psi}_c \overline{\phi}_b \Phi_{b c a_3 \cdots a_N}.$$

Exchanging ψ and ϕ is equivalent to exchanging the indices b and c on Φ, which changes the sign, so $\{a(\psi), a(\phi)\} = 0$. Taking the Hermitian conjugate gives $\{a^\dagger(\psi), a^\dagger(\phi)\} = 0$. In particular $a^\dagger(\psi)^2 = 0$: we can't put two fermions in the same state.

For the anticommutator of $a(\psi)$ and $a^\dagger(\phi)$ we again act on an N-particle tensor $A_{a_1 \cdots a_N}$, which is now antisymmetric

$$\begin{aligned}(a(\psi)a^\dagger(\phi)A)_{a_1 \cdots a_N} &= (N+1)\overline{\psi}_b \phi_{[b} A_{a_1 \cdots a_N]} \\ (a^\dagger(\phi)a(\psi)A)_{a_1 \cdots a_N} &= N\overline{\psi}_b \phi_{[a_1} A_{|b|a_2 \cdots a_N]}.\end{aligned}$$

As in the bosonic case, we split the $N+1$ terms in the first antisymmetrization according to which index sits on ϕ. The term with b there gives $\phi_b A_{a_1 \cdots a_N}$. Placing a_j there instead requires the transposition $b \leftrightarrow a_j$, which carries a minus sign, and leaves b in the jth slot of A. Moving b to the front of A costs a further $(-1)^{j-1}$, which is exactly the sign that a_j carries in the antisymmetrization over $a_1 \cdots a_N$ in the second line. Hence

$$(N+1)\overline{\psi}_b \phi_{[b} A_{a_1 \cdots a_N]} = \overline{\psi}_b \phi_b A_{a_1 \cdots a_N} - N\overline{\psi}_b \phi_{[a_1} A_{|b|a_2 \cdots a_N]}.$$

The relative minus sign is the only difference from the bosonic case, and it turns the commutator into an anticommutator

$$\{a(\psi), a^\dagger(\phi)\} = \overline{\psi}_b \phi_b = \langle \psi | \phi \rangle.$$

Taking ψ and ϕ to be position eigenstates gives the fermionic field operator relations Equation 1.19.

4.2 Alternative Derivation of One- and Two-particle Operators

Here we give simpler derivation of the general form of one- and two- particle operators using the form Equation 1.17 for the action of the field operator $\psi(x)$, which applies in both the bosonic and fermionic cases

4.2.1 One particle operators

We show that the one-particle operator $\hat{A}$ in Equation 3.1 has the representation

$$\hat{A} = \int \psi^\dagger(x_1)[A_{x_1} \psi(x_1)]\, dx_1$$

where the subscript on A_x is to indicate that the operator acts on the field operator $\psi(x)$ as if it were a wavefunction e.g. the momentum operator acts as $p\psi(x) = -id\psi(x)/dx$.

We now evaluate an arbitrary matrix element of the operator between states $|\Psi\rangle$ and $|\Phi\rangle$ (of any form, not necessarily product states) using Equation 1.17

$$\begin{aligned}\langle\Psi|\hat{A}|\Phi\rangle &= \int \langle\Psi|\psi^\dagger(x_1)[A_{x_1}\psi(x_1)]|\Phi\rangle\, dx_1 \\ &= N\int \Psi^*(x_1, x_2, \ldots x_N)\left[A_{x_1}\Phi(x_1, x_2, \ldots x_N)\right] dx_1 dx_2 \ldots dx_N \\ &= \sum_{j=1}^{N}\int \Psi^*(x_1, x_2, \ldots x_N)\left[A_{x_j}\Phi(x_1, x_2, \ldots x_N)\right] dx_1 dx_2 \ldots dx_N.\end{aligned}$$

The last line is just the matrix element of Equation 3.1 in the first quantized representation. Note that all terms in the sum over j are identical because we can permute the arguments of Ψ and Φ to bring the argument x_j to the first position. Any sign in the fermionic case is squared and therefore has no effect.

4.2.2 Two particle operators

We show that the general two particle operator Equation 3.7 has representation

$$\hat{B} = \frac{1}{2}\int \psi^\dagger(x_1)\psi^\dagger(x_2)[B_{x_1,x_2}\psi(x_2)\psi(x_1)]\, dx_1 dx_2,$$

where B_{x_1,x_2} is an operator that acts on a pair of wavefunctions. Taking matrix elements and using Equation 1.17 again yields

$$\begin{aligned}\langle\Psi|\hat{B}|\Phi\rangle &= \frac{1}{2}\int \langle\Psi|\psi^\dagger(x_1)\psi^\dagger(x_2)[B_{x_1,x_2}\psi(x_2)\psi(x_1)]|\Phi\rangle\, dx_1 dx_2 \\ &= \frac{1}{2}N(N-1)\int \Psi^*(x_1, x_2, \ldots x_N)\left[B_{x_1,x_2}\Phi(x_1, x_2, \ldots x_N)\right] dx_1 dx_2 \ldots dx_N \\ &= \sum_{j<k}\int \Psi^*(x_1, x_2, \ldots x_N)\left[B_{x_j,x_k}\Phi(x_1, x_2, \ldots x_N)\right] dx_1 dx_2 \ldots dx_N.\end{aligned}$$

The last line is the first quantized representation of the matrix elements of Equation 3.7. Again all terms in the sum over pairs are identical by (anti-)symmetry of the wavefunctions.

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